

DAY 4 WEEK 1

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PROJECT: Battery SoC Estimation & Remaining Charging-Time
Prediction Using Machine Learning

Day 4: Dataset Design & Quality Expectations

1. Dataset — Structure

Field	Role	Type	Meaning
Battery Voltage	Input	Numerical	Voltage level
Charging Current	Input	Numerical	Current supplied
Charging Power	Input	Numerical	Power used
Battery Temperature	Input	Numerical	Temperature effect
Charging History / Session ID	Input / identifier	Mixed	Previous/session information
State of Charge	Target	Numerical	Charge percentage

(SoC)			
Remaining	Target	Numerical	Time left to full charge
Charging Time			

Target Outputs

- State of Charge (SoC) — Percentage of battery charge.
- Remaining Charging Time — Estimated time left for full charging.

Data Splitting

- Data will be divided by charging session, rather than individual rows.
- This helps prevent data leakage and gives a fair model evaluation.

2. Data-Related Challenges

- **Missing Sensor Data** —
- Some voltage, current, or temperature readings may be missing.

- **Noisy Measurements** —
 - Sensor readings may contain errors or sudden fluctuations.
- **Data Leakage** —
 - Data from the same charging session should not appear in both training and testing.
- **Unequal Session Lengths** —
 - Charging sessions may start, stop, or have different durations.
- **Imbalanced SoC Data** —
 - Some battery charge levels may have more samples than others.
- **Redundant Features** —
 - Some features may provide similar information, such as charging power calculated from voltage and current.
- **Outliers** —
 - Unusual voltage, current, or temperature values can affect model predictions.